

Machine Learning-Based Prediction of Final Batting Score in ICC Cricket World Cup Matches

Katuri Manikanta Pavankalyan

Roll No: 241FA18214

Vignan's Foundation for Science, Technology & Research

Department of Artificial Intelligence and Machine Learning

Abstract—Cricket is a data-rich sport in which information such as runs, wickets, overs, teams, and venue is generated throughout a match. This paper presents a machine learning-based approach for predicting the final batting score of a team during an innings. The dataset used in this work contains 21,031 records. The input features include year, city, Team 1, Team 2, batting team, bowling team, over number, runs scored in an over, runs scored till that over, wickets taken in an over, wickets taken till that over, current run rate, and remaining overs.

Data preprocessing includes checking missing values and duplicate records, creating the target variable, feature engineering, and categorical encoding. A match-wise group-based 80–20 train-test split is used to prevent records from the same match from appearing in both training and testing data. Five regression algorithms are evaluated: Linear Regression, Decision Tree, Random Forest, Gradient Boosting, and Extra Trees.

The models are evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 . Among the tested models, Extra Trees provides the best overall performance with an MAE of 43.64, MSE of 3337.32, RMSE of 57.77, and R^2 of 0.3094. A sample unseen test observation with an actual final score of 286 runs was predicted as 284.7 runs.

Index Terms—Cricket analytics, machine learning, regression, score prediction, Extra Trees, sports analytics.

I. INTRODUCTION

Cricket produces a large amount of structured information during every match. The number of overs completed, runs scored, wickets taken, batting team, bowling team, and venue provide useful information about the current state of an innings.

Predicting the final score from the current state of an innings is a regression problem because the output is a continuous numerical value. Machine learning algorithms can learn relationships between innings progress and final score from historical cricket data.

The objective of this project is to develop and compare machine learning regression models for predicting the final batting score in ICC Cricket World Cup matches.

The major objectives are:

- preprocess and understand the cricket dataset;
- create useful innings-progress features;
- encode categorical variables;
- split the data into training and testing sets;
- train multiple regression models;
- compare model performance using standard metrics;
- select the best-performing model; and
- perform prediction on unseen test data.

II. DATASET DESCRIPTION

The dataset contains 21,031 records and information related to cricket matches and innings progress.

The original dataset contains 20 columns. Important variables include Year, Match Number, City, Team 1, Team 2, Batting Team, Bowling Team, Over Number, Runs Scored in Over, Runs Scored Till That Over, Wicket Taken in Over, and Wickets Taken Till That Over.

The target variable is `final_score`, representing the final score of the team that batted first.

A. Feature Set

The final model uses 13 input features:

III. DATA PREPROCESSING

The preprocessing stage consists of several steps.

A. Missing Values

Missing values were checked using the Pandas `isnull().sum()` operation. No missing values were found, so missing-value imputation was not required.

B. Duplicate Records

Duplicate records were checked using `duplicated().sum()`. The result was zero, indicating that no duplicate rows were present.

TABLE I
FINAL INPUT FEATURES

Feature	Type
Year	Numerical
City	Categorical
Team 1	Categorical
Team 2	Categorical
Batting Team	Categorical
Bowling Team	Categorical
Over Number	Numerical
Runs Scored in Over	Numerical
Runs Scored Till That Over	Numerical
Wicket Taken in Over	Numerical
Wickets Taken Till That Over	Numerical
Current Run Rate	Numerical
Remaining Overs	Numerical

C. Target Variable

The target variable `final_score` was created using the final score of the team that batted first in each match.

D. Current Run Rate

Current Run Rate (CRR) was created to represent the scoring rate of the batting team:

$$CRR = \frac{\text{Runs Scored Till That Over}}{\text{Over Number}} \quad (1)$$

For example, if a team has scored 60 runs after 10 overs, the current run rate is 6 runs per over.

E. Remaining Overs

The remaining number of overs was calculated as:

$$\text{RemainingOvers} = 50 - \text{OverNumber} \quad (2)$$

These two engineered features increase the final feature set from 11 to 13 features.

F. Categorical Encoding

Categorical variables such as City, Team 1, Team 2, Batting Team, and Bowling Team were converted into numerical representations so that they could be processed by the regression algorithms.

IV. TRAINING AND TESTING

Because several observations can belong to the same cricket match, a normal random row-wise split can cause data from one match to appear in both training and testing datasets.

To avoid this problem, a unique match group was created using Year and Match Number. A

TABLE II
TRAINING AND TESTING DATA

Dataset	Shape
X_{train}	(16870, 13)
X_{test}	(4161, 13)
y_{train}	(16870,)
y_{test}	(4161,)

group-based 80–20 split was then performed using `GroupShuffleSplit`.

The resulting datasets were:

Approximately 80% of the data is therefore used for training and 20% for testing.

V. MACHINE LEARNING MODELS

Five regression algorithms were trained and evaluated.

A. Linear Regression

Linear Regression was used as a baseline model. It attempts to model the relationship between input variables and the final score using a linear function.

B. Decision Tree Regression

Decision Tree Regression uses a sequence of decision rules to divide the feature space and predict a numerical target.

C. Random Forest Regression

Random Forest Regression combines multiple decision trees and averages their predictions to produce the final prediction.

D. Gradient Boosting Regression

Gradient Boosting builds trees sequentially, where subsequent trees attempt to reduce the errors of earlier trees.

E. Extra Trees Regression

Extra Trees Regression is an ensemble learning method that constructs multiple randomized decision trees and combines their predictions.

VI. EVALUATION METRICS

The models were evaluated using four metrics.

A. Mean Absolute Error

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3)$$

Lower MAE indicates smaller average prediction error.

TABLE III
MODEL PERFORMANCE

Model	MAE	MSE	RMSE	R^2
Linear Regression	47.26	3701.13	60.84	0.2340
Decision Tree	69.11	7808.07	88.36	-0.6160
Random Forest	51.37	4457.66	66.77	0.0780
Gradient Boosting	45.80	3447.68	58.72	0.2870
Extra Trees	43.64	3337.32	57.77	0.3094

B. Mean Squared Error

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (4)$$

Lower MSE is preferred.

C. Root Mean Squared Error

$$RMSE = \sqrt{MSE} \quad (5)$$

Lower RMSE indicates better prediction performance.

D. R-squared

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (6)$$

A higher R^2 generally indicates that the model explains a larger portion of the variation in the target.

VII. RESULTS AND DISCUSSION

The five models were evaluated using the same test dataset.

Extra Trees achieved the lowest MAE, MSE, and RMSE among the five models and also achieved the highest R^2 value.

Therefore, Extra Trees was selected as the final model.

It is important to note that the R^2 value of 0.3094 is not directly comparable with classification accuracy. This is a regression problem, and the model is predicting a numerical cricket score. The obtained R^2 value reflects the predictive performance of the selected features and dataset.

VIII. VISUALIZATION

Model evaluation can be visualized using MAE, MSE, RMSE, and R^2 comparisons. These visualizations help compare the performance of the five regression models.

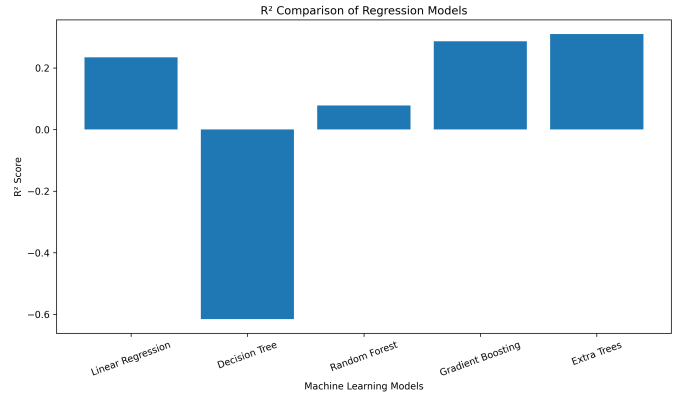


Fig. 1. R^2 comparison of regression models.

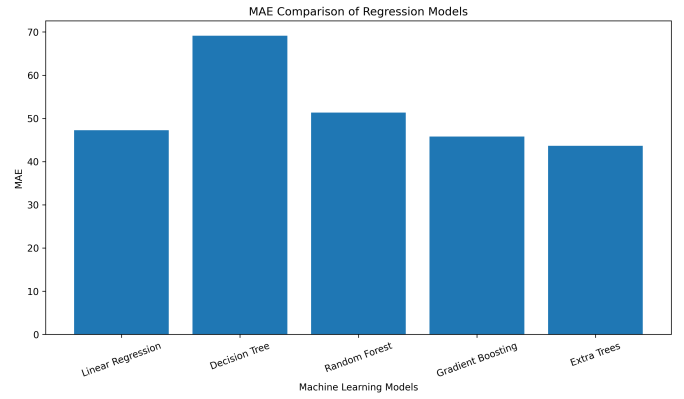


Fig. 2. MAE comparison of regression models.

IX. SAMPLE PREDICTION

The selected Extra Trees model was used to predict the final score for an unseen test observation.

The recorded prediction was:

The absolute prediction error is:

$$|286 - 284.7| = 1.3 \quad (7)$$

This demonstrates that the model can produce a close prediction for an individual unseen test observation. However, the overall model performance is represented by the evaluation metrics calculated over the complete test set.

X. DISCUSSION

The experimental results indicate that Extra Trees performed better than the other tested regression models for the selected features.

However, the obtained R^2 value is approximately 0.31. This does not mean that the project failed. Final

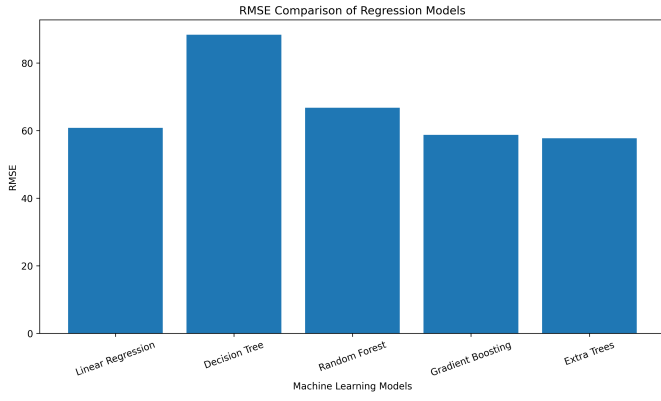


Fig. 3. RMSE comparison of regression models.

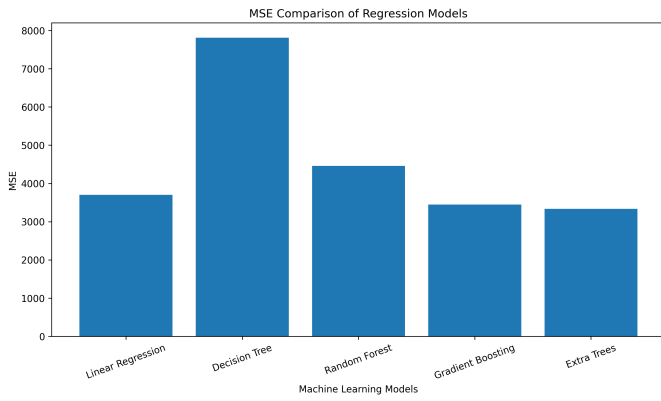


Fig. 4. MSE comparison of regression models.

cricket scores are influenced by many factors, including player performance, pitch conditions, match situations, and other information that may not be represented in the current dataset.

The results therefore represent the performance obtained using the available features rather than an artificially optimized score.

The comparison also demonstrates why multiple regression algorithms should be evaluated rather than selecting a model based on a single metric or assumption.

XI. CONCLUSION

This paper presented a machine learning-based approach for predicting the final batting score in ICC Cricket World Cup matches.

The project used 21,031 records and 13 final input features. Data preprocessing included missing-value checking, duplicate checking, target creation, feature engineering, and categorical encoding. A match-wise

TABLE IV
SAMPLE PREDICTION

Parameter	Score
Actual Final Score	286
Predicted Final Score	284.7
Absolute Error	1.3

group-based 80–20 split was used to separate training and testing data.

Five regression algorithms were trained and compared. Extra Trees achieved the best performance among the tested models with an MAE of 43.64, MSE of 3337.32, RMSE of 57.77, and R^2 of 0.3094.

A sample unseen observation with an actual score of 286 runs was predicted as 284.7 runs, resulting in an absolute error of 1.3 runs.

Future work can improve the prediction system by incorporating additional information such as player-level statistics, pitch conditions, weather, recent player form, and detailed ball-by-ball information.

REFERENCES

- [1] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [2] P. Geurts, D. Ernst, and L. Wehenkel, “Extremely randomized trees,” *Machine Learning*, vol. 63, no. 1, pp. 3–42, 2006.
- [3] J. H. Friedman, “Greedy function approximation: A gradient boosting machine,” *The Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.
- [4] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [5] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. New York, NY, USA: Springer, 2009.
- [6] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer, 2006.
- [7] K. P. Murphy, *Machine Learning: A Probabilistic Perspective*. Cambridge, MA, USA: MIT Press, 2012.
- [8] W. McKinney, “Data structures for statistical computing in Python,” in *Proc. 9th Python in Science Conf.*, 2010, pp. 51–56.
- [9] C. R. Harris et al., “Array programming with NumPy,” *Nature*, vol. 585, pp. 357–362, 2020.
- [10] J. D. Hunter, “Matplotlib: A 2D graphics environment,” *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007.
- [11] R. Kohavi, “A study of cross-validation and bootstrap for accuracy estimation and model selection,” in *Proc. 14th Int. Joint Conf. Artificial Intelligence*, 1995, pp. 1137–1143.
- [12] G. James, D. Witten, T. Hastie, and R. Tibshirani, *An Introduction to Statistical Learning: With Applications in R*. New York, NY, USA: Springer, 2013.